Language Models for Analysing Social Media Discourse: Resources and Applications

Thesis submitted in accordance with the requirements of Cardiff University for the degree of Doctor in Philosophy by

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Declaration

I hereby declare that, this thesis (and the work presented in it) is submitted to the School of Computer Science & Informatics, Cardiff University, in fulfilment of the requirements for the degree of Doctor of Philosophy, and that this thesis is entirely my own work.

Cardiff, ADD DATE

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Abstract

Social media has become an integral part of daily life for millions worldwide, serving as a platform where people exchange opinions, express concerns, and influence one another. From natural disasters to political debates, social media platforms like X (formerly Twitter) and Reddit have become primary sources of news and public discourse. This activity results in the generation of vast amounts of data each day which, due to their sheer volume and largely unstructured nature, researchers and analysts face significant challenges in effectively studying and analysing it.

To analyse this complex data source, researchers frequently utilise Natural Language Processing (NLP) tools and techniques. However, social media data poses unique challenges distinct from more formal or structured text. The landscape of social media is constantly evolving, with new slang, trending topics, emojis, and brevity of posts creating additional hurdles that traditional NLP methods often struggle to address.

The emergence of Large Language Models (LLMs), such as GPT-3 (Brown et al., 2020a), has transformed the landscape of natural language processing, making it possible to analyse and generate text at an unprecedented scale. However, our findings indicate that smaller, specialized models can often perform competitively with, and occasionally outperform, these larger models on domain-specific tasks. significant when considering the resource and environmental costs associated with training and deploying large-scale models, underscoring the importance of efficient, task-oriented model development.

In this work, we introduce several domain-specific models and datasets tailored specifically for social media data. We apply these specialized tools to studies focusing on politically charged content, hate speech detection, and misinformation. By leveraging these resources, we aim to demonstrate that specialized models not only offer competitive performance but also provide a more sustainable approach to social media data analysis.

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Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Pretraining Approach
ELMo	Embeddings from Language Models
BoW	Bag of Words
ICL	in-context learning
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
LM	Language Model
MLM	Masked Language Model

PLM	Pre-trained Language Model
ML	Machine Learning
NLP	Natural Language Processing
SoTA	State of The Art
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
URL	Uniform Resource Locator
MTLD	Measure of Textual Lexical Diversity
GloVe	Global Vectors for Word Representation
T5	Text-to-Text Transfer Transformer
BART	Bidirectional and Auto-Regressive Transformers
Flan-T5	Fine-tuned Language Net T5
C4	Colossal Clean Crawled Corpus
QLoRA	Quantized Low-Rank Adaptation
NER	Named Entity Recognition
LLaMA	Large Language Model Meta AI
NMF	Non-negative Matrix Factorization

1 Introduction

The rise of social media platforms have transformed the way people communicate and interact online. Platforms like *Twitter* and *Reddit*, have grown to be an integral part of the lives of millions of people, with approximately 63% of the population using a social media platform daily [1], and are seen as key sources of real-time information, public opinion, and social dynamics, making them valuable data sources for understanding human behavior and societal trends (Barberá, 2015; Shu et al., 2017). As populations shape opinions and influence each other through these platforms (Ausat, 2023), it has become crucial, from both business and scientific perspectives, to better understand how people interact and what factors drive their behaviors online.

This need becomes even more apparent in the context of political discourse, given how easily voters can be influenced and manipulated through targeted social media campaigns Goodwin et al. (2023). For example, Zhang et al. (2025) demonstrate how President Trump's often incendiary posts on Twitter and Truth Social amplified social media user engagement and attracted increased news media

[1] https://www.statista.com/statistics/617136/digital-population-worldwide/

attention. Similar studies on other major political events, such as Brexit (Mora-Cantallops et al., 2021), highlight the pivotal role social media platforms play in shaping how information spreads through networks, facilitating content generation, and influencing user behaviour.

To effectively leverage this rich but chaotic data, NLP provides an essential set of tools for extracting, structuring, and analyzing textual content from social media. Standard tasks include sentiment analysis(Zimbra et al., 2018), topic modeling (Egger and Yu, 2022), and entity recognition (Nasar et al., 2021), which help quantify emotional tone, uncover underlying themes, and identify key actors within discussions. Recent advances in neural architectures, such as transformers (Subakan et al., 2021), enable deeper contextual understanding and have paved the way for more nuanced tasks, including stance detection (AlDayel and Magdy, 2021), misinformation tracking (Hayawi et al., 2022), and aspect-based sentiment analysis (Brauwers and Frasincar, 2022). These methods allow researchers to convert raw, unstructured posts into structured formats that support downstream analysis.

Furthermore, social media data allows for tracking of user behaviour, discourse evolution, and community dynamics (Feuerriegel et al., 2025; Arazzi et al., 2023). Through the use of NLP techniques it becomes possible to detect coordinated campaigns, assess the spread of narratives, or evaluate public reactions over time (Luceri et al., 2024). These insights are increasingly valuable in fields such as computational social science, marketing, and policy analysis, where understanding the connections between language, behaviour, and influence is crucial. The intersection of NLP and social media offers significant opportunities for both methodological innovation and impactful, real-world applications.

Despite their potential, analysing social media data poses numerous challenges for

the field of NLP. User-generated content on these platforms often lacks context and is highly informal, containing slang, emojis, abbreviations, typos, and other forms of noisy data (Eisenstein, 2013; Gain et al., 2023). Moreover, language usage on social media evolves rapidly, making it difficult for static models and lexicons to keep pace. These characteristics demand robust, adaptive, and context-aware approaches to effectively process and interpret the vast and dynamic stream of social media text.

In order to tackle the limitations above, we explore existing methodologies and resources that work on understanding complex social interactions on social media. Subsequently, we introduce new models and datasets that improve on previous work, apply them on large corpora collected from *Twitter* and *Reddit* to understand better human interactions regarding politics discourse, and misinformation and hate speech proliferation. The rest of this introductory chapter is organised as follows: § 1.1 outlines the research aim and motivations for this thesis, § 1.2 outlines the contribution of research, § 1.3 lists publications that are peer-reviewed or currently under review, and § 1.6 describes the structure of the thesis and summarises this introductory chapter.

1.1 Motivation

NLP applications are essential for understanding human interactions in social media, especially in the context of political discourse and dissemination of misinformation and hate speech. For example, Machine Learning (ML) models are used by social platforms, such as Facebook (Schroepfer, 2021), to detect harmful content and protect users from online abuse. Similarly, researchers have developed tools to analyse political communication on Twitter (Barberá, 2015), track the spread of misinformation (Shu et al., 2017), and understand the dynamics of online discussions (Zhang et al., 2018).

Many hate speech detection models rely on supervised learning techniques, which require large amounts of annotated data to train effectively. However, the availability of high-quality annotated datasets for hate speech detection is limited, and existing datasets often suffer from biases and inconsistencies (Davidson et al., 2017). This can lead to models that are not generalisable across different contexts or languages, limiting their effectiveness in real-world applications. Other applications, such as misinformation detection and sentiment analysis, also face similar challenges in terms of data availability and quality. For example, misinformation detection models often rely on large-scale datasets of social media posts, which can be difficult to obtain and may not accurately reflect the diversity of online content (Shu et al., 2017). Similarly, sentiment analysis models often struggle to capture the nuances of social media language, which can vary significantly across different platforms and user demographics (Barbieri et al., 2022).

Due to the heavily unstructured content often shared in social media content, many of the aforementioned applications rely on specialised datasets to perform well. These datasets are often collected from social media platforms, such as *Twitter* and *Reddit*, and annotated by human coders to provide ground truth labels for training and evaluation. However, the quality and reliability of these datasets can vary significantly, depending on the annotation process and the expertise of the annotators (Klie et al., 2024). This can lead to biases and inaccuracies in the models trained on these datasets, limiting their generalisability and effectiveness in real-world scenarios.

The proliferation of social media platforms has created unprecedented challenges in political discourse and misinformation spread (Muhammed T and Mathew, 2022). Political polarization and the rapid dissemination of false information have become major societal concerns, affecting democratic processes and public opinion formation (Allcott and Gentzkow, 2017). These challenges are further complicated by the heavily unstructured nature of social media content, which can pose difficulties even for larger models (Eisenstein, 2013; Gain et al., 2023). Finally, it is crucial to recognise both the potential biases in existing models, tools, and datasets, as well as the environmental impact associated with developing and deploying LLMs (Bender et al., 2021).

1.2 Contributions

The main motivation of this thesis is to explore how NLP techniques can be effectively applied to social media data in order to address complex societal questions. Social media platforms generate vast amounts of unstructured textual

data that offer unique insights into public opinion, communication patterns, and societal events. However, leveraging this data poses significant challenges due to its informal language, high variability, and noise. This raises a central research question: How can NLP methodologies be adapted or developed to robustly analyse and interpret social media content for meaningful societal insights? This question branches into more specific problems concerning data representation, model design, annotation strategies, and evaluation practices.

Rather than attempting to solve all aspects of the problem, this thesis aims to fill gaps in existing literature on concrete subproblems where progress can be made on how to detect subtle patterns in public discourse, how to model social dynamics from textual evidence, and how to evaluate such models in realistic settings. Specifically, the research presented in this thesis focuses on two core areas: (1) developing and evaluating NLP resources for social media analysis; and (2) applying these resources through well-defined approaches to analyse social media dynamics in the context of politics discourse, misinformation spread and hate speech propagation. By addressing these areas, the thesis aims to provide new insights into the dynamics of social media communication and contribute to the development of more effective tools for understanding and combating harmful online behaviours. To this end, the thesis makes a number of noticeable contributions as listed below:

1.2.1 Part 1: Modeling Approaches for Social Media Language Understanding

The first core contribution of the thesis focuses on developing and evaluating NLP tools specifically designed for social media applications.

I.1 **New Datasets for Topic Classification on Twitter.** Extracting meaningful insights from the vast amount of data generated on Twitter is inherently challenging, and performing targeted topic analysis adds further complexity. Unsupervised methods such as topic modelling are often difficult to evaluate and interpret, while existing datasets for supervised classification are not well-suited for the informal and dynamic nature of social media content. To address this gap, we introduce two novel datasets, TweetTopic and it is multilingual extension X-Topic, specifically designed for training and evaluating topic classification models on Twitter data. These datasets, along with corresponding State of The Art (SoTA) models, are detailed in Chapters **??** and 3.

I.2 **Unification of Hate Speech Detection Datasets.** We present a unified hate speech detection dataset specifically curated for social media content by leveraging and combining multiple existing resources. Although hate speech detection is a widely studied problem, many available datasets lack diversity and tend to focus on specific target groups, limiting the generalisability of trained models. By systematically aggregating and harmonising these resources, we construct a comprehensive dataset that, as demonstrated in **??**, enables models to achieve superior performance compared to those trained on individual datasets.

I.3 **Introduction of a Benchmark for Twitter-related Tasks.** We present SuperTweetEval, a benchmark that unifies and standardises the evaluation of social media NLP tasks, addressing the current fragmentation in this area. Comprising XX diverse tasks, our benchmark poses significant challenges not only for smaller Pre-trained Language Models (PLMs) but also for the latest LLMs, as discussed in **??**.

A common observation across all three contributions is that smaller, specialised models, such as RoBERTa, can perform as well as, and often outperform, larger LLMs like GPT-4 on the evaluated tasks when fine-tuned. This observation is primarily due to the specialised training data: the models we evaluated are frequently trained on large corpora of social media content, making them better suited for domain-specific language use and nuances. Additionally, their architectural design contributes to this advantage. Unlike generative LLMs, which are based on a decoder architecture, smaller PLMs typically employ an encoder architecture, which is inherently more effective for discriminative tasks such as classification. Encoders focus on learning contextual representations of input sequences, which allows for precise feature extraction and robust classification performance which is particularly important for tasks like topic classification and hate speech detection, where subtle linguistic cues and contextual dependencies are critical.

The impact of this thesis is evidenced by the widespread adoption of the released datasets and models. For instance, the topic classification resources introduced in Chapters **??** and 3 have accumulated over 1,200,000 downloads on the Hugging Face platform[2], while our hate speech classifier (**??**) has more than 1,400,000

[2] https://huggingface.co/collections/cardiffnlp/tweettopic-65eb2a0eada92

downloads[3], highlighting their practical value and community uptake.

Apart from the wide adoption of the resources, the models and datasets have been used in various studies from different domains. For example, they have informed research in tourism studies, where Lee et al. (2025) employed our topic classification model to perform a targeted investigation on online hostility directed at tourists in the context of over-tourism debates. Similarly, our hate speech classifier (**??**) has demonstrated performance comparable to, or surpassing, established tools such as Google's Perspective API[4] and OpenAI's Moderation system[5], as shown by Shen et al. (2025). It has also been adopted by researchers to examine the temporal dynamics of hate speech on Twitter Arun et al. (2024), and to investigate correlations between hate speech and crime statistics (Bicakci et al., 2025).

This interdisciplinary application underscores the utility of robust NLP tools in social science and humanities contexts, as evidenced by citations our papers have achieved across computer science, linguistics, political science, psychology, business, environmental science, medicine, and philosophy[6].

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[3] https://huggingface.co/cardiffnlp/twitter-roberta-base-hate-latest/settings

[4] https://perspectiveapi.com/

[5] https://platform.openai.com/docs/guides/moderation

[6] According to data from https://www.semanticscholar.org/

1.2.2 Part 2: Analysing Harmful and Political Dynamics in Social Media

We use the resources developed and insights extracted in the previous chapters to analyse complex social media dynamics.

II.1 **Politics and Negativity.** We conduct a large-scale analysis of posts by politicians from eight different parliaments across five languages (including English, Spanish, and Greek) to investigate how negatively charged tweets tend to spread more widely and rapidly than neutral or positive ones. To achieve this, we train domain-specific sentiment analysis models and develop a systematic approach to quantify and track the diffusion of such content. Further details are provided in **??**.

II.2 **Profiling Misinformation Spreaders.** We present a methodology to profile users who frequently share misinformation on social media. By leveraging our previously developed models for topic classification and hate speech detection, along with sentiment and emotion analysis, we systematically analyse a large number of user accounts to uncover behavioural patterns and content characteristics associated with misinformation spread. Further details are provided in **??**.

II.3 **Trigger Points Analysis.** Trigger points are a relatively new concept referring to specific words or phrases that can provoke strong emotional reactions in discourse. We propose an approach for systematically identifying potential trigger points using various lexical features, focusing mainly on sentiment, emotion, and the presence of hate speech. Unlike previous

qualitative studies that introduced the notion of trigger points, our approach adopts a quantitative perspective. Further details are provided in **??**.

Throughout this work, we provide detailed and transparent methodologies and approaches that are not only robust but also easy to replicate and adapt. The approaches we introduce are designed to be easily applicable to other datasets and platforms beyond Twitter and Reddit, enabling researchers and practitioners to extend our analyses to a wide range of social media sources and related contexts.

Finally, it is worth noting that there was a consistent effort to collaborate with experts from other fields, such as social sciences and healthcare, and to incorporate their feedback in order to address real-world problems 'through our research. These interdisciplinary collaborations, as seen in § 1.3.2.1, highlight the practical relevance of our work.

1.3 Publications

The majority of the material in this thesis have been published in peer-reviewed conferences and journals, while the remaining material are under reviewed at peer-reviewed conferences. Below is a list of the core publications where the main contributions of the thesis have been published in the chronological order.

1.3.1 Core Publications

Below is a list of the core publications where the main contributions of the thesis have been published in chronological order. The majority of these the publications have been led by myself. However, in two cases, pecifically publications (1) and (7), the second co-author made substantial contributions. In publication (1), Asahi Ushio was responsible for collecting the raw data and designing the annotation guidelines used to create the TweetTopic dataset (**??**). In publication (7), Christian Arnold developed the statistical framework employed to examine the effect of trigger points (**??**).

1.3.1.1 Part 1: Modeling Approaches for Social Media Language Understanding

1. Dimosthenis Antypas, Asahi Ushio, Jose Camacho-Collados, Vitor Silva, Leonardo Neves, and Francesco Barbieri. 2022. Twitter Topic Classification. In Proceedings of the 29th International Conference on Computational Linguistics, pages 3386–3400, Gyeongju, Republic of Korea. International Committee on Computational Linguistics. **??**

2. Dimosthenis Antypas, Asahi Ushio, Francesco Barbieri, and Jose Camacho-Collados. 2024. Multilingual Topic Classification in X: Dataset and Analysis. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 20136–20152, Miami, Florida, USA. Association for Computational Linguistics. Chapter 3

3. Dimosthenis Antypas and Jose Camacho-Collados. 2023. Robust Hate Speech

Detection in Social Media: A Cross-Dataset Empirical Evaluation. In The 7th Workshop on Online Abuse and Harms (WOAH), pages 231–242, Toronto, Canada. Association for Computational Linguistics. **??**

4. Dimosthenis Antypas, Asahi Ushio, Francesco Barbieri, Leonardo Neves, Kiamehr Rezaee, Luis Espinosa-Anke, Jiaxin Pei, and Jose Camacho-Collados. 2023. SuperTweetEval: A Challenging, Unified and Heterogeneous Benchmark for Social Media NLP Research. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 12590–12607, Singapore. Association for Computational Linguistics. **??**

1.3.1.2 Part 2: Analysing Harmful and Political Dynamics in Social Media

5. Dimosthenis Antypas, Alun Preece, and Jose Camacho-Collados. 2023. Negativity spreads faster: A large-scale multilingual Twitter analysis on the role of sentiment in political communication. Online Social Networks and Media, 33:100242. **??**

6. Dimosthenis Antypas, Alun Preece, and Jose Camacho-Collados. 2024. A Multi-Faceted NLP Analysis of Misinformation Spreaders in Twitter. In Proceedings of the 14th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis, pages 71–83, Bangkok, Thailand. Association for Computational Linguistics. **??**

7. Dimosthenis Antypas, Christian Arnold, Jose Camacho-Collados, Nedjma Ousidhoum, and Carla Perez Almendros. 2024. Words as Trigger Points in Social Media Discussions. *Under Review in ICSWM 2025.* **??**

1.3.2 Other Publications

The following publications are not formally part of this thesis. However, they illustrate the applicability of the research in diverse real-world scenarios, particularly in handling the nuances of social media data. They also highlight the interdisciplinary relevance of our work as they connect NLP with social sciences, health care, and disaster management.

1.3.2.1 Collaboration with public stakeholders

During the duration of this thesis there were opportunities to collaborate with various public stakeholders addressing challenges arising from social media usage. These collaborations resulted in several publications, showcasing the real-world applicability of our methods.

The first paper (8) emerged from a collaboration with the Spanish National Office Against Hate Crimes. It's aim was to examine potential temporal trends in hate speech on Twitter, particularly around Spanish national holidays (e.g., Labour Day). My role focused on adapting our existing hate speech detection models to achieve robust performance in the Spanish language.

The second paper (9) was produced in collaboration with Public Health Wales. This work aimed to develop an approach for early detection of influenza outbreaks using Twitter data. My primary contribution involved curating suitable datasets and training RoBERTa-based models capable of identifying tweets that reported influenza-like symptoms, either from a first-person perspective or in general.

Another collaboration (10) included colleagues from the School of Earth and Environmental Sciences and aimed to provide post-disaster recovery assessment in the case of earthquakes by utilising sentiment analysis on relevant tweets. Using our expertise on the area we managed to train tailored and domain specific classifiers which out performed the initial, paid, service that was being used for sentiment analysis.

Finally, a collaboration with researchers from Snap Inc. (11) resulted in the development and public release of a dataset specifically designed for the detection of sensitive content on social media, along with accompanying detection models. My primary contributions to this project involved coordinating the collection and aggregation of annotations, as well as training and evaluating the Language Models (LMs) employed in the study.

8. Jesús Gómez, Alberto Matilla-Molina, Ma Pilar Amado, Dimosthenis Antypas, Jose Camacho-Collados, Carlos J. Máñez, Tomás Fernández-Villazala, Alicia Méndez-Sanchís, and Javier López. 2023. The interaction between offensive and hate speech on Twitter and relevant social events in Spain. In News Media and Hate Speech Promotion in Mediterranean Countries, pages 81–109. IGI Global.

9. Mark Drakesmith, Dimosthenis Antypas, Claire Brown, Jose Camacho-Collados, and Jiao Song. 2025. Towards a social media-based disease surveillance system for early detection of influenza-like illnesses: A Twitter case study in Wales. In Proceedings of the Tenth Workshop on Noisy and User-generated Text (W-NUT 2025).

10. Diana Contreras, Dimosthenis Antypas, Javier Hervas, Sean Wilkinson, Jose

Camacho-Collados, Philippe Garnier, and Cécile Cornou. 2025. Post-disaster recovery assessment using sentiment analysis of English-language tweets: A tenth-anniversary case study of the 2010 Haiti earthquake. Sustainability 17, no. 11: 4967.

11. Dimosthenis Antypas, Indira Sen, Carla Perez-Almendros, Jose Camacho-Collados, and Francesco Barbieri. 2025. Sensitive Content Classification in Social Media: A Holistic Resource and Evaluation. In The 9th Workshop on Online Abuse and Harms (WOAH), Vienna, Austria. Association for Computational Linguistics.

1.3.2.2 Complementary Studies on Social Media

In addition to collaborations with public stakeholders, several projects were undertaken that, while adjacent to the main goals of this thesis, are not part of its core contributions. Specifically, in work (12), we conducted a large-scale analysis of Twitter data, collecting a substantial corpus of tweets related to the COVID-19 pandemic. From this, we extracted and annotated a labelled dataset aimed at identifying pandemic-related misinformation and developed binary classifiers for misinformation detection.

Work (13) focused on classifying Twitter posts by geographic origin (European vs. non-European) and deriving insights from these groups using comparative techniques such as lexical specificity and word embedding models. My main contribution to this study involved the analysis of the collected corpus, particularly through the use of lexical specificity metrics (Camacho-Collados et al., 2016).

Work (14) utilised Reddit data, and aimed to develop classifiers capable of detecting early signs of depression in user posts. My involvement centered on exploring model architectures capable of handling long-form textual input, which was a significant challenge at the time.

Finally, work (15) was part of a large collaborative effort that resulted in the development of the TweetNLP Python library. This toolkit facilitates working with Twitter data by offering easy-to-use implementations for tasks such as sentiment analysis, topic classification, and question answering. My contributions primarily involved integrating models I had previously developed, such as those for topic classification and hate speech detection, into the library.

12. Dimosthenis Antypas, Jose Camacho-Collados, Alun Preece, and David Rogers. 2021. COVID-19 and Misinformation: A Large-Scale Lexical Analysis on Twitter. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing: Student Research Workshop, pages 119–126, Online. Association for Computational Linguistics.

13. David Tuxworth, Dimosthenis Antypas, Luis Espinosa-Anke, Jose Camacho-Collados, Alun Preece, and David Rogers. 2021. Deriving disinformation insights from geolocalized Twitter callouts. In Proceedings of KDD 2021 - Workshop on Deriving Insights From User-Generated Text.

14. David Owen, Dimosthenis Antypas, Athanasios Hassoulas, Antonio F. Pardiñas, Luis Espinosa-Anke, and Jose Camacho Collados. 2023. Enabling early health care intervention by detecting depression in users of web-based forums using language models: Longitudinal analysis and evaluation. JMIR

AI 2, no. 1.

15. Jose Camacho-collados, Kiamehr Rezaee, Talayeh Riahi, Asahi Ushio, Daniel Loureiro, Dimosthenis Antypas, Joanne Boisson, Luis Espinosa Anke, Fangyu Liu, and Eugenio Martínez Cámara. 2022. TweetNLP: Cutting-Edge Natural Language Processing for Social Media. In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pages 38–49, Abu Dhabi, UAE. Association for Computational Linguistics.

1.3.2.3 Other work

The previous work discussed has been largely adjacent to the core focus of this thesis, particularly with an emphasis on social media. In addition to this, I contributed to other projects that are less directly related. Specifically, in project (16), I was involved in the development of BLEnD, a benchmark designed to evaluate the cultural diversity of LLMs. This was achieved by introducing manually annotated question pairs covering everyday cultural knowledge from 16 countries and in 13 languages. My role focused on collecting and ensuring the quality of the Greek questions. In project (17), a large-scale study investigating how LMs process metaphors and analogies, I contributed to the creation of parts of the datasets used for model evaluation.

16. Myung, Junho, Nayeon Lee, Yi Zhou, Jiho Jin, Rifki Afina Putri, Dimosthenis Antypas, Hsuvas Borkakoty et al. "BLEnD: A Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages." *Under Review in NeurIPS 2024 Datasets and Benchmarks Track.*

17. Joanne Boisson, Asahi Ushio, Hsuvas Borkakoty, Kiamehr Rezaee, Dimosthenis Antypas, Zara Siddique, Nina White, and Jose Camacho-Collados. 2024. How Are Metaphors Processed by Language Models? The Case of Analogies. In Proceedings of the 28th Conference on Computational Natural Language Learning, pages 365–387, Miami, FL, USA. Association for Computational Linguistics.

1.4 Ethics and Privacy

The research included in the thesis has been approved by Cardiff University's ethics comitee and also follows the guidelines of ACL code of ethics[7].

Specifically, as our research focuses on socia media we have taken caution while collecting data used in our experiments to adhere to the aforementioned ethic rules. Firstly, all the data were collected using the official Application Programming Interfaces (APIs) provided by *Twitter* and *Reddit*. Moreover, the privacy of the users whose data is used is protected by masking user handles and Uniform Resource Locators (URLs) to prevent any potential identification of users. Exceptions are data colleted from verified[8] *Twitter* users which are already public figures such as politicians, celebrites, and goverment employees.

Finally, regarding the annotations conducted, all coders were paid at least minimum wage for their work, and no personal identifying information was collected. This applies to the manual data labeling for TweetTopicand X-Topic,

[7]https://www.acm.org/code-of-ethics

[8]As defined by the legacy verified policy of Twitter in https://help.x.com/en/managing-your-account/legacy-verification-policy.

as described in Chapters **??** and 3 respectively.

1.5 Naming conventions

Models. In Chapters **??** and **??**, the models used are described as SoTA. While this statement may not hold true at the time of writing this thesis, they were the best-performing models when the experiments were conducted.

Twitter. A lot of the data used were collected from X while it was known as *Twitter* through the use of it's now defunct academic API. In the thesis we will refer to the platform as *Twitter* to reflect the time of the data collection and research done. Exception being Chapter 3 where the data was collected from X and we will refer to the platform as X.

1.6 Thesis Outline and Summary

In this section, we provide an overview of the structure of the thesis and summarise the core themes of each chapter. Figure 1.1 shows the entire structure of this thesis, where chapters with same interest are grouped together and chapters that utilise tools are colored coded the same way.

The first set of chapters are preliminary chapters to review the literature of relational representation learning and LMs (Chapters 1 and 2). The second set of chapters focus on identifying limitations in existing resources related to social media NLP and proposing solutions. Mainly we present our study for topic

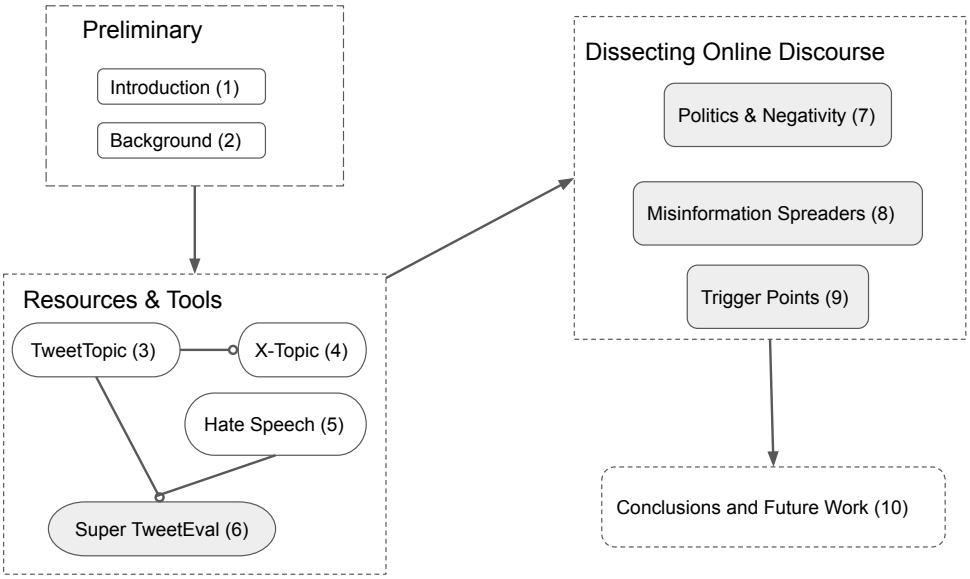

Figure 1.1: Outline of the thesis. The chapter number is indicated in brackets. The grey color indicates connections between chapters of Part 1 and Part 2.

classification in *Twitter* (**??**), and our multilingual extension to this work (Chapter 3). Following this we introduce an ensemble of models and datasets related to hate speech detection (**??**), and a benchmark for social media NLP tasks (**??**) which builds upon our previous work.

The final part of the thesis focuses on applying the aforementioned resources to understand complex social interactions in social media. **??** introduces a study on the role of sentiment in political communication, followed by an analysis of misinformation spreaders (**??**), and finally we investigate how specific words can act as trigger points in online discussions (**??**). Finally, we conclude the thesis with a summary of the contributions and future work (**??**).

As a temporal reference for this thesis, Figure 1.2 presents a high-level timeline of

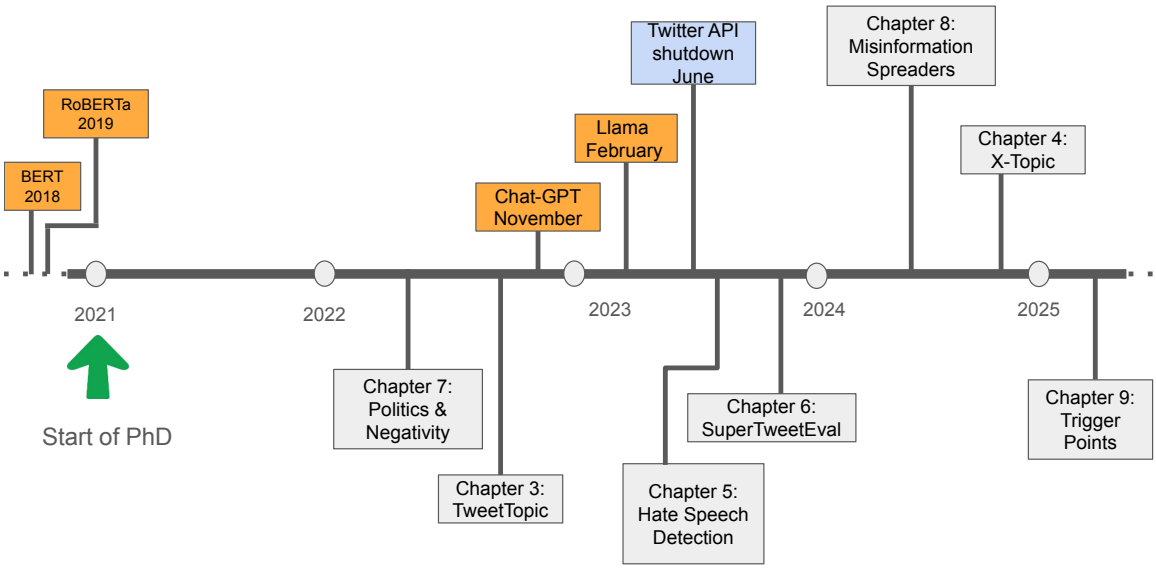

Figure 1.2: Timeline of papers publications along with major events in NLP field.

the publications that form part of this work, along with major events in the field of NLP, such as the release of key models and significant developments like the shutdown of the Twitter API. Although this thesis primarily focuses on utilising smaller LMs, the work conducted after the release of ChatGPT and open-source models like LLaMA incorporates results from these models and often compares their performance as well.

In the next chapter (Chapter 2), we will discuss the relevant literature and methodologies related to the research presented in this thesis. We will review existing work on key topics such as sentiment analysis, hate speech detection, misinformation analysis, and topic classification, and provide a brief, high-level introduction to computational models like LMs.

2 Background

This chapter provides the theoretical and methodological foundation of the thesis, covering the evolution of NLP tools and their applications to social media data. It explores the challenges of processing noisy, heterogeneous text from platforms like Twitter and discusses existing work on key areas such as hate speech detection, misinformation analysis, and multilingual topic classification. By situating the thesis within the broader research landscape, this chapter demonstrates how the presented contributions advance the state of the art.

2.1 Language Models

LMs are fundamental to NLP, aiming to capture the statistical properties of language and predict the probability of a sequence of words. Their evolution has significantly impacted the capabilities of NLP systems, especially in handling complex human language.

2.1.1 Traditional Language Models

2.1.1.1 Early Language Models

Early LMs were often based on statistical methods, such as N-gram models, which predict the next word based on the previous N-1 words (Buck et al., 2014). These models, while foundational, suffered from limitations in capturing long-range dependencies and semantic meaning. To represent words, early systems frequently relied on static word embeddings such as Word2Vec (Mikolov et al., 2013) and Global Vectors for Word Representation (GloVe) (Pennington et al., 2014), where each word was mapped to a single, fixed vector regardless of context. While these representations improved over one-hot encodings by capturing some semantic similarity (e.g., "king" and "queen" being close in vector space), they failed to reflect how a word's meaning could shift depending on its usage in different contexts.

These limitations were overcomed, up to a degree, through the development of neural approaches capable of modeling sequential dependencies and context more effectively. Specifically, Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTMs) and Gated Recurrent Units (GRUs), introduced a new class of language models capable of processing word sequences while maintaining a dynamic internal state (Yu et al., 2019; Dey and Salem, 2017). These models improved the contextual understanding of language over static embeddings by updating representations as each token in a sequence was processed. However, they still faced challenges in handling long-term dependencies and lacked efficient parallelization during training, which restricted scalability and performance

on large datasets (Sherstinsky, 2020).

2.1.1.2 Masked Language Models

Masked Language Models (MLMs) emerged with the introduction of the Transformer architecture (Subakan et al., 2021). Unlike traditional LMs that predict the next word, MLMs are trained by masking a percentage of the tokens in the input and then predicting those masked tokens based on the context provided by the unmasked tokens as seen in Figure 2.1 (Devlin et al., 2019). This bidirectional training allows them to learn richer, contextualized representations of words. These models are typically pretrained on massive corpora such as Wikipedia and the Common Crawl[1], enabling them to learn robust language representations.

For example, the embedding for the word bank in "he sat by the river bank" will differ significantly from that in "she went to the bank to deposit money," because the surrounding context informs the model about the intended sense of the word. This is in stark contrast to earlier approaches like Word2Vec or GloVe, which assign the same vector to bank regardless of its meaning in different contexts. These contextual embeddings allow MLMs to capture intricate semantic and syntactic relationships, making them highly effective across a wide range of downstream tasks.

Most contemporary MLMs are based on the BERT architecture, which is an encoder-based model leveraging bidirectional self-attention mechanisms to capture contextual information from both the left and right of each token (Devlin et al., 2019). Extensions to BERT include Robustly Optimized BERT Pretraining

[1] https://commoncrawl.org

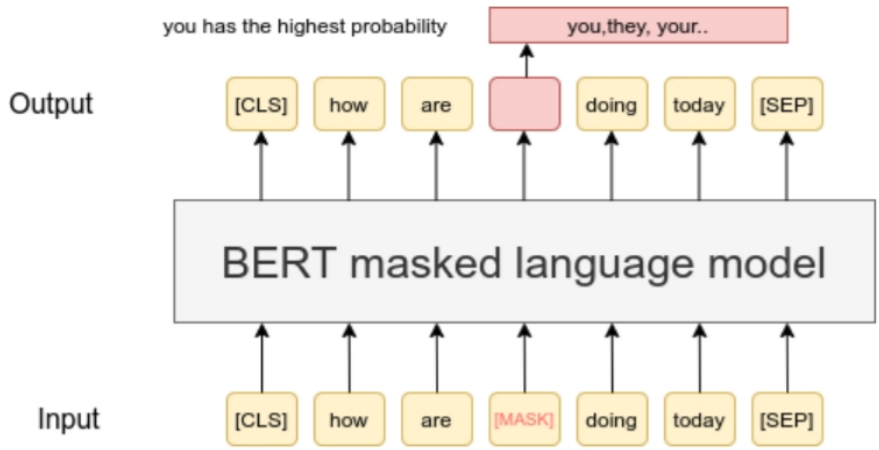

Figure 2.1: Example of MLM objective for training in Bidirectional Encoder Representations from Transformers (BERT).

Approach (RoBERTa), which optimises the pre-training process and introduces larger datasets, longer training, and XLM-R, which extends RoBERTa to multilingual settings (Liu et al., 2019; Conneau et al., 2020).

One of the main benefits of these models is that their knowledge can be utilised in various downstream tasks with minimal modifications. Also, due to their encoder architecture which allows them to understand context from both left and right, they display significantly improved performance when fine-tuned on tasks like text classification, question answering, and named entity recognition (Yang et al., 2023).MLMs have become the backbone of many state-of-the-art NLP systems, enabling significant advancements in understanding and generating human language. They are particularly effective for tasks that require deep contextual understanding, such as sentiment analysis (Alaparthi and Mishra, 2021), hate speech detection (Mozafari et al., 2019), and misinformation detection (Kaliyar et al., 2021).

2.1.2 Encoder-Decoder Models

Encoder-decoder models represent a distinct approach in language modeling, particularly effective for sequence-to-sequence tasks. These architectures consist of two components: (i) an encoder that processes the input sequence and compresses its information into a latent representation, and (ii) a decoder that generates an output sequence based on this representation. This architecture is well-suited for tasks where the output differs in form or length from the input, such as translation, summarization, and question generation. Similar to MLMs, encoder-decoder models are also be pretrained on large corpora, while leveraging self-attention mechanisms in both the encoder and decoder stacks which allows them to learn bidirectional representations in the encoder while simultaneously training the decoder to generate sequences from noisy or incomplete inputs.

Encoder-decoder models became popular with the introduction of Text-to-Text Transfer Transformer (T5) (Raffel et al., 2020a). T5 casts NLP problems, such as summarisation, machine translation, and question asnwering, into a unified text-to-text format. It is pretrained on the Colossal Clean Crawled Corpus (C4) corpus(Raffel et al., 2020b) using a span corruption objective, where spans of text are masked and replaced with unique sentinel tokens, and the model learns to reconstruct the missing spans which enables it to develop robust representations for both understanding and generation (Figure 2.2).

An extension of T5, Fine-tuned Language Net T5 (Flan-T5) (Chung et al., 2024), further improves performance through instruction tuning (Wei et al., 2021) on a wide variety of tasks and datasets which enhances it's zero-shot and few-shot generalization capabilities. Another notable encoder-decoder model is Bidirectional

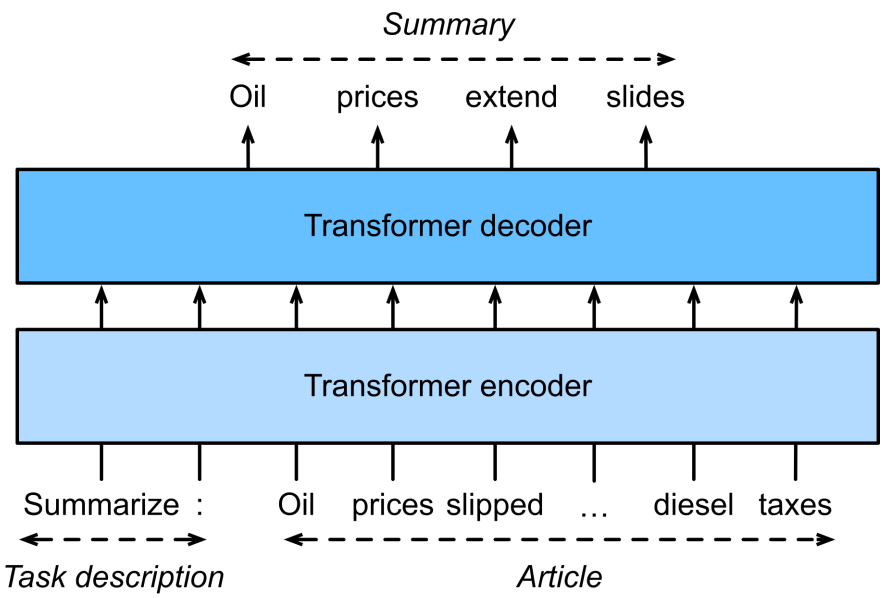

Figure 2.2: Example of T5 finetuning for a summarisation task.

and Auto-Regressive Transformers (BART) (Lewis et al., 2019), which is trained on a denoising autoencoder objectivewhere the model learns to reconstruct original text from corrupted input. BART combines the strengths of both MLMs and autoregressive models, making it effective for tasks like summarisation and translation.

Encoder-decoder models are widely used in tasks where the input and output differ significantly in structure or length, and similarly to MLMs they can be further finetuned for specific applications. Common use cases include machine translation, text summarization, question answering, and question generation. They can also be adapted for text generation tasks, producing coherent and contextually relevant text from prompts. However, these models are generally more complex and computationally intensive than MLMs, and they not always outperform encoder-only models on tasks that do not involve sequence generation (Gasparetto

et al., 2022). At the same time, they often lag behind decoder-only models, such as LLMs, in open-ended text generation tasks (Saattrup Nielsen et al., 2025).

2.1.3 Large Language Models

LLMs represent the most recent advancement in language modeling, building upon the foundations laid by MLMs and encoder-decoder models. They are characterized by their massive scale, often containing hundreds of billions of parameters, and their ability to perform a wide range of tasks with remarkable performamce, often in a zero-shot or few-shot learning setting (Wang et al., 2020). Unlike traditional LMs, which are typically trained on specific tasks or datasets, LLMs are pretrained on vast and diverse datasets encompassing text, code, and other modalities like images.

Architecturally, most LLMs are based on the Transformer decoder-only design and are trained using an autoregressive objective (Brown et al., 2020a)where, given a sequence of tokens, the model learns to predict the next token in a left-to-right manner. In contrast to bidirectional models like BERT, decoder-only models do not have access to future tokens during training, which makes them suited for generative tasks such as text completion, dialogue, summarization, and code generation.

Modern LLMs are capable to perform tasks based purely on natural language instructions, making *prompt engineering* a critical aspect of their effective use (Reynolds and McDonell, 2021). Moreover, similarly to smaller LMs their performance on specific tasks can often be further improved through finetuning. This is typically applied to smaller versions of the models using techniques like

Quantized Low-Rank Adaptation (QLoRA) (Dettmers et al., 2023), which enables efficient adaptation with limited computational resources.

Several notable LLMs have significantly advanced the field of NLP. For example, OpenAI's GPT series, starting with GPT-2 (Radford et al., 2019), demonstrated the potential of large-scale unsupervised pretraining and autoregressive text generation. Its succesors, GPT-3 (Brown et al., 2020b) and GPT-4 (Achiam et al., 2023) further scaled up the model size and training data, achieving impressive performance on a wide range of tasks with few-shot and zero-shot learning capabilities.

Alongside proprietary models, several open-sourced LLMs have been introduceded. Meta's Large Language Model Meta AI (LLaMA) family (Touvron et al., 2023a,b) offers models that are computationally efficient and accessible to the research community, with versions as small as 1 billion parameters. Other prominent open-source alternatives include Mistral 7B and Mixtral (Jiang et al., 2024)—a mixture-of-experts model using sparse attention to activate only a subset of its parameters during inference—and Qwen (Bai et al., 2023), a high-performing model widely adopted in academia due to its open availability and strong benchmark results.

These models showcase that decoder-only architectures when scaled massively are able to achieve broad generalization across NLP tasks with minimal supervision.

However, despite their impressive capabilities, LLMs pose significant challenges. These include high computational costs, issues related to interpretability, and the risk of generating biased or harmful content (Ferrara, 2023; George et al., 2023). While decoder-only LLMs excel in generative and open-ended tasks, encoder-only

models, such as those based on the BERT architecture, remain highly competitive and often superior for certain structured tasks, like text classification and Named Entity Recognition (NER), where their bidirectional context is advantageous. Moreover, encoder-based models tend to be more efficient to train and deploy for such applications.

2.2 Social media platforms

Social media platforms have become central to the study of natural language in real world environments, offering vast amounts of real-time, user-generated content. These platforms differ significantly from traditional text sources such as books, news articles, or formal documents. Their informal, dynamic, and highly contextual nature presents unique opportunities and challenges for NLP research and applications (Mustofa and Saptomo, 2025). A lot of these paltforms focus on different modalities, for example, Instagram emphasises photos and videos, while others like Snapchat are inherently multimodal. In contrast, Twitter and Reddit have emerged as particularly rich sources of linguistic data and are widely used in computational social science and NLP research (). This is primarily because text is the dominant medium of communication on these platforms, and they used to offer relatively easy access to data through official APIs [2]. This thesis focuses on these two platforms due to their complementary structures and modes of interaction.

Twitter is structured around short, standalone messages called "tweets," which are constrained to 280 characters. Although tweets are restricted in length, they often convey substantial information and sentiment, enhanced by the use of hashtags,

[2] Twitter's Academic API shut down on XXX. Reddit's API shutdown on XXX

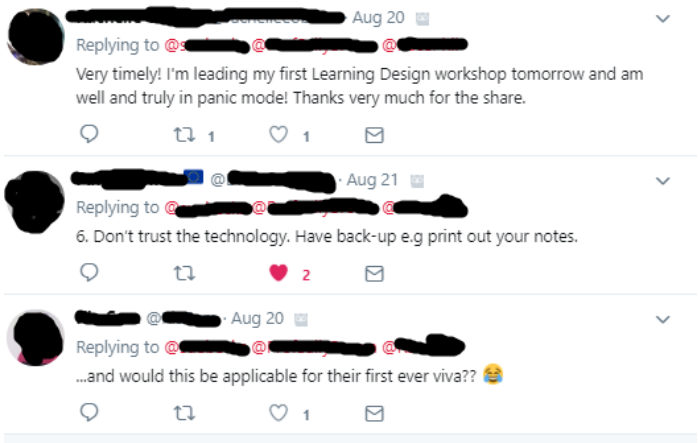

Figure 2.3: An example of a Twitter post and replies to it.

mentions, emojis, and embedded media. User interactions take the form of replies, retweets, and quote tweets, which together create loosely connected conversational threads (Figure 2.3). The character limit encourages users to be linguistically inventive, resulting in the widespread use of slang, abbreviations, and non-standard grammatical constructions.

Reddit, in contrast, is a forum-based platform organized into topic-specific communities known as *subreddits*. Posts on Reddit can be significantly longer than tweets and often initiate threaded discussions that allow for deeper, hierarchical conversations (2.4). This structure offers richer context and more coherent discourse patterns.

Although Reddit and Twitter differ in format and interaction design, they display several common linguistic characteristics as both are informal, rapidly evolving, and rely on similar engagement mechanisms such as upvotes and likes. These features shape the type and tone of user-generated content, making both platforms valuable for studying opinion formation, discourse dynamics, and online social behaviour.

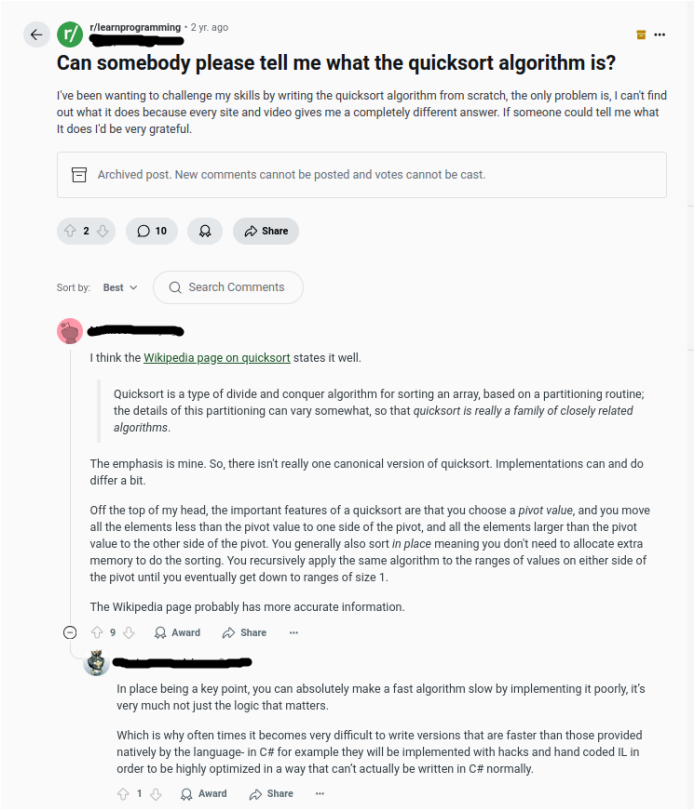

Figure 2.4: An example of a Reddit thread from r/learningprogramming subreddit.

2.2.1 Specialised Language Models for Social Media

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The direct application of LMs trained on formal or curated text corpora often fails to generalize effectively to social media data. Social media text presents a wide range of linguistic challenges for NLP due to its informal and dynamic nature. It often features slang, abbreviations (e.g., "brb", "imo"), and non-standard grammar, which deviate from conventional language norms. Posts frequently lack sufficient context, leading to ambiguity in interpretation, particularly when sarcasm or irony

is involved. Additionally, platform-specific metadata such as emojis, hashtags, and mentions play an important role in conveying meaning and identifying topics. Multilinguality and code-switching are also common, with users often blending languages or dialects within a single post. Language on these platforms evolves rapidly, with new slang and trends emerging constantly, requiring models to adapt over time. Social media content also reflects societal biases and may include toxic or harmful language, posing ethical and technical challenges. Finally, the short and often noisy nature of posts, particularly on platforms like Twitter, often requires careful pre-processing or tailored model architectures.

These challenges have led to the development of specialised language models that are pre-trained or fine-tuned on large-scale social media corpora to capture the idiosyncrasies of online discourse (Nguyen et al., 2020).

These models are explicitly designed to handle the noise and variability inherent in social media and they often incorporate strategies for addressing spelling errors, abbreviations, emojis, and platform-specific syntax (e.g., hashtags, mentions, retweet markers). By learning from in-domain data, these models are better equipped to understand the linguistic norms, stylistic features, and emergent expressions characteristic of social platforms.

Popular examples of specialised language models include: (i) BERTweet (Nguyen et al., 2020), a RoBERTa-based model trained exclusively on 850M English tweets; (ii) Twitter-RoBERTa (Loureiro et al., 2022), which continues training from the RoBERTa checkpoint using a large tweet corpora; (iii) Bernice (DeLucia et al., 2022a) a multilingual social media model designed to handle cross-lingual and code-switched content; and (iv) DialoGPT (Zhang et al., 2019) A conversational response generation model built by fine-tuning GPT-2 on approximately 147 million

Reddit comment exchanges. These examples reflect a common trend: RoBERTa is the backbone architecture for many social media-specific models. This is largely due to its robustness, relatively low computational requirements for large-scale pretraining, and the fact that its main limitation, a short maximum context length of 512 tokens, is not a major drawback for short-form social media text. Specialized models trained on domain-specific social media data consistently outperform their general-purpose counterparts and, in some tasks, match or surpass LLMs in downstream applications such as hate speech detection, stance classification, and misinformation detection ().

2.3 NLP Applications for Social Media

In this section, we explore core NLP tasks applied to social media data and that are significant for this thesis, highlighting their importance and the unique challenges they present. These applications include sentiment analysis, emotion detection, topic classification, hate speech detection, and misinformation analysis (Table 2.1). Each of them plays a crucial role in understanding user-generated content and extracting valuable insights from the vast amount of data available on social media platforms.

2.3.1 Sentiment Analysis

Sentiment analysis aims to determine the emotional tone or overall sentiment expressed in a piece of text. It is commonly framed as a multi-class classification task, where a model classifies a given text as negative, neutral, or positive based on

Task	Post	Label
Sentiment Analysis	Just got a promotion at work! Feeling on top of the world #Blessed	Positive
Emotion Detection	Can't believe he forgot my bday AGAIN... smh	Anger
Hate Speech Detection	Go back to where u came from. Ppl like u don't belong here	Hate Speech
Topic Classification	The new iPhone cam is insane but battery still trash lol #TechTalk	Technology

Table 2.1: Examples of social media posts across different classification tasks.

its sentiment (Mejova, 2009). Alternative approaches include aspect-based sentiment analysis, where the model assesses the sentiment of a text towards a specific target or entity (Zhang et al., 2022). Sentiment analysis can also be modeled as an ordinal regression task, where the model assigns a score on a predefined scale (e.g. 0 to 5), with 0 representing the most negative and 5 the most positive sentiment (Saad and Yang, 2019).

On social media, sentiment analysis is crucial for understanding public opinion toward brands, products, events, or political figures. For example, Matalon et al. (2021) applied sentiment analysis to examine opinion inversion in political tweets related to the Israeli-Palestinian conflict. Similarly, Rodríguez-Ibáñez et al. (2021) analyzed Twitter posts mentioning Spanish political parties to study sentiment dynamics over time. Alaparthi and Mishra (2021) used sentiment analysis to investigate public attitudes toward COVID-19 vaccines on Twitter.

Other researchers have explored the use of sentiment analysis in brand monitoring and customer feedback. Chowdhury (2024) proposed a framework for analysing customer reviews on social media to help businesses understand customer satisfaction and improve their products or services. Similarly, Chamlertwat et al.

(2012) demonstrated how sentiment analysis can be used to extract consumer insights from social media data, enabling companies to adapt their strategies based on public opinion.

Sentiment analysis has also been employed to assess public reactions to major events, such as natural disasters or political elections. For instance, Willson et al. (2021) used sentiment analysis to examine Twitter reactions to the 2019–2020 Australian bushfires, revealing regional and temporal variations in sentiment. Similarly, Behl et al. (2021) applied sentiment analysis to tweets related to the COVID-19 pandemic, helping identify user concerns and needs during the crisis and illustrating how social media can inform disaster response strategies.

In this thesis, sentiment analysis, framed as a multi-class problem, is used extensively to extract insights into users' opinions and attitudes on various topics discussed on social media. This is particularly prominent in Chapters **??**, **??**, and **??**.

2.3.2 Emotion Detection

Emotion detection while similar to sentiment analysis it is more complex and challenging task as it aims to identify more granular human emotions rather than overall negative, neutral, and positive emotional state. Specifically, an emotion detection model provided a text will highlight specific emotions like anger, joy, sadness, fear, surprise, disgust, etc that are present in it. Emotion detection can be framed as a multi-class classification task (Balabantaray et al., 2012), where each class corresponds to a specific emotion, or as a multi-label classification task (Zhang et al., 2020), where multiple emotions can be detected simultaneously in a

single text. It can also be approached as an ordinal regression task, where the model assigns a score to each emotion (e.g. 0-5) based on its intensity or relevance to the text (Muhammad et al., 2025).

Emotion detection is important for social media analysis particularly relevant when there is need for more fine-grained understanding of emotional responses to events or content on social media. Researchers have utilised emotion detection to study political discourse, marketing campaigns, and hate speech studies among others. For example Bhaumik et al. (2023) apply emotion detection techniques on Twitter posts related to the 2017 French presidential elections and study the effect of emotional content as influence indicators in online political discussions.Researchers (Rahib et al., 2022), have also used emotion detection to explore the effect of the COVID-19 pandemic to the general population and study how they are responding to and dealing with evolving circumstances. Finally, Schäfer and Kistner (2023) investigate the connection between hate speech and emotions and propose a method for jointly modeling both phenomena.

In this thesis, emotion detection, framed as multi-label task, is utilised in Chapters **??**, and **??** as part of our analysis to extract insights from users about the emotional status.

2.3.3 Topic Modelling

Topic modelling is focuses on identifying the underlying themes or topics present in a collection of texts. It is often framed as an unsupervised learning task, where the model discovers latent topics without predefined labels. Common approaches include Latent Dirichlet Allocation (LDA) (Blei et al., 2003), Non-negative Matrix

Factorization (NMF) (Lee and Seung, 1999), and more recent methods like BERT-based topic models such as BertTopic (Grootendorst, 2022) and Top2Vec (Angelov, 2020). These methods aim to group similar documents based on their content, allowing for the identification of prevalent themes within a corpus.

The unsupervised nature of topic modelling while beneficial in many cases, can also be a limitation when specific topics or categories are of interest. In such cases, supervised topic classification is often used, where a model is trained on labeled data to predict predefined topics for new texts. This approach is particularly useful when performing targeted analysis where specific specific categories, such as politics, technology, and sports are of interest. This is particular important on social media, as it helps in organizing vast amounts of information, identifying trending discussions that are of interest and filter content based on specific themes.Applications of topic classification on social media focus on identifying trending topics (Lee et al., 2011; Bouadjenek et al., 2022) but have also been utilised often in parallel with sentiment analysis to gain insights into public opinion and discourse (Huang et al., 2013; Rodrigues and Chiplunkar, 2022).

In this thesis we utilise topic classification in Chapters **??**, **??**,**??** with the aim of community profiling and identifying topics of interest for specific groups of users.

2.3.4 Hate Speech Detection

Hate speech detection involves identifying offensive, threatening, or discriminatory language. It is a critical application of NLP on social media, aiming to identify content that expresses hatred or encourages violence towards individuals or groups based on attributes like race, ethnicity, religion, gender, and sexual orientation.

This is a highly sensitive and challenging task that is crucial for moderation and creating safer online environments. Hate speech detection can be framed as a binary classification task, where the model predicts whether a given text contains hate speech or not (Saleh et al., 2023) but can an also be approached as a multi-class classification task where the model identifies specific types of hate speech (e.g., racism, sexism, homophobia) (Qureshi and Sabih, 2021), or as a multi-label classification task, where multiple forms of hate speech can be present in a single text (Dirting et al., 2022).

Hate speech detection is particularly important on social media platforms, where the rapid spread of harmful content can have significant real-world consequences as was the case during the Rohingya crisis in Myanmar (Nourooz Pour, 2023). Researchers have applied hate speech detection to various social media platforms, including Twitter and Reddit, to study the prevalence and impact of hate speech in online discourse. For example, Kim and Kesari (2021) investigates relations between misinformation and hate speech specifically in the case of the COVID-19 pandemic and hate speech target to Asian communities. Other researchers (Agarwal et al., 2021) have explore the nature and prevalence of hate speech in political dialogue in Twitter and also sutdies that focus on specific expressions of hate speech like misogyny and abuse of female politicians on Twitter (Fuchs and Schäfer, 2021).

In this thesis, hate speech detection is framed as a multi-class classification task and is used in Chapters **??**, **??**, and **??** to identify harmful content and understand the dynamics of online hate speech.

2.3.5 Misinformation & Polarisation

WIP

Misinformation and disinformation analysis is a critical area of NLP research, particularly in the context of social media platforms where the rapid spread of false or misleading information can have significant societal impacts. The spread of misinformation on social media contributes to public harm, especially in contexts like health communication and political discourse. Identifying false or misleading content is complicated by subtle rhetorical strategies and the rapid evolution of narratives.

Misinformation analysis includes tasks such as misinformation detection, stance detection, propaganda detection, source credibility assessment, and linguistic feature analysis. These tasks are essential for understanding how misinformation spreads, its impact on public opinion, and how to counteract it through fact-checking and verification efforts. Misinformation detection involves identifying false or misleading content, often framed as a binary classification task where the model predicts whether a given text contains misinformation or not (). Stance detection, on the other hand, determines whether a text agrees, disagrees, discusses, or is unrelated to a given claim, and can be framed as a multi-class classification task (). Propaganda detection focuses on identifying rhetorical techniques used to manipulate public opinion, often framed as a multi-class classification task where the model identifies specific propaganda techniques used in a text (). Source credibility assessment involves evaluating the reliability of the source of information, which can be framed as a binary classification task where the model predicts whether a source is credible or not (). Finally, linguistic feature analysis involves

analyzing linguistic patterns (e.g., sensationalist language, emotional intensity, specific vocabulary) often associated with misinformation, which can be framed as a feature extraction task where the model identifies specific linguistic features in a text ().

On the other hand polarisation analysis focuses on understanding the strong division of the populace in two or more sides (), and the formation of echo chambers and selective exposure, where users are exposed only to information that aligns with their existing beliefs. Polarisation often leads to the reinforcement of biases, the spread of misinformation, and the deterioration of constructive discourse (). Tasks related to polarisation can include echo chamber detection, opinion dynamics tracking, stance and ideology detection, linguistic markers of polarisation, and studying disagreement and deliberation in online discussions. Echo Chambers and Filter Bubbles: NLP can identify communities with shared beliefs and analyze how information flows within and between these groups. Opinion Dynamics: Tracking how opinions evolve over time and how they are influenced by social interactions and exposure to different narratives. Linguistic Markers of Polarisation: Analyzing specific language choices (e.g., divisive rhetoric, "us vs. them" framing, emotional language) that indicate or contribute to polarization. Disagreement and Deliberation: Studying how online discussions devolve into unproductive arguments versus constructive dialogue.

In this thesis we look at both misinformation and polarisation effects, specifically in Chapter **??** by investigating user profiles likely to share misinformation, and in Chapters **??** **??** focuses on polarisation in social discourse.

Part I

Modeling Approaches for Social Media Language Understanding

3

Multilingual Topic Classification in X: Dataset and Analysis

In the previous chapter (**??**), we introduced the task of tweet topic classification, presented an English-language dataset, and evaluated several transformer-based models. In this chapter, we expand the scope of the task to include multiple languages. We introduce X-Topic, a multilingual dataset featuring content in four distinct languages (English, Spanish, Japanese, and Greek), crafted for the purpose of tweet topic classification. Our dataset includes a wide range of topics, tailored for social media content, making it a valuable resource for scientists and professionals working on cross-linguistic analysis, the development of robust multilingual models, and computational scientists studying online dialogue. Finally, we leverage X-Topic to perform a comprehensive cross-linguistic and multilingual analysis, and compare the capabilities of current general- and domain-specific language models.

3.1 Introduction

Social platforms such as X , Snapchat and Instagram provide an environment for content creation and information sharing among people and organisations. In particular, people use these platforms to express their sentiments, share their opinions on multiple topics, and discuss and influence each other (Barbieri et al., 2014; Hu et al., 2021; Ansari et al., 2020). In this scenario, these platforms are rich sources for informal short text, as they include content about recent events, shared by a heterogeneous group of users. The vast amount of content shared on social media, however, make it impossible to analyse and digest it without automatic tools.

Unsupervised approaches such as Latent Dirichlet Allocation (LDA) (Blei et al., 2003) and topic modelling variations (Steyvers and Griffiths, 2007), or more recently, BERTopic (Grootendorst, 2022), are common approaches to deal with this issue. However, these methods are usually built as an ad-hoc analysis, with the derived topics not being easily interpretable or comparable among different analyses. On the other hand, when looking at supervising approaches, existing resources mainly focus on the news articles domain, e.g., BBC News (Greene and Cunningham, 2006), Reuter (Lewis et al., 2004), 20News (Lang, 1995), and WMT News Crawl (Lazaridou et al., 2021) with few exceptions like scientific (arXiv) (Lazaridou et al., 2021) and medical (Ohsumed) (Hersh et al., 1994) domains.

This chapter focuses on expanding the resources available for multilingual tweet classification. We leverage an initial topic taxonomy of 19 topics, first proposed in Antypas et al. (2022), and introduce the new X-Topic dataset that includes tweets

from four different languages: English, Spanish, Japanese and Greek. Our dataset is focused on X data and aims to address the lack of labelled multilingual social media data, as well as to encourage the creation of new methods for multilingual topic classification.

By leveraging X-Topic as a benchmark, we explore multiple model architectures and sizes for multilingual tweet topic classification: (1) zero-shot, (2) few-shot, (3) monolingual, (4) cross-lingual and (5) multilingual. Our analysis highlights the challenging nature of the task and reveals interesting patterns in relation to the use of LLMs and supervised approaches for the topic classification task in social media, especially in relation to the type of data considered for training.

The X-Topic dataset, as well as the topic classification models built upon it, are made openly available. X-Topic is available at `https://huggingface.co/datasets/cardiffnlp/tweet_topic_multilingual`. Table 3.1 shows some sample instances of X-Topic for each language. Finally, the best multilingual models of *base* and *large* sizes are available at `https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base-topic-multilingual` and `https://huggingface.co/cardiffnlp/twitter-xlm-roberta-large-topic-multilingual`, respectively.

3.2 Related Work

The task of classifying topics in social media content has garnered significant attention from the research community in recent years (Schlichtkrull et al., 2023;

Tweet	Topics
en: I don't think I really want to go to Coachella unless Taylor Swift is headlining	Celebrity & Pop Culture, Music
es: quiero una date en un museo **translation**: I want a date in a museum	*Relationships, Arts & Culture, Diaries & Daily Life*
ja: 久々になーーんもしないでいい日が二日もあるのでゆっくり富平井絆果と向き合うよ **translation**: It's been a long time since I've had two days where I don't have to do anything, so I'm going to take my time and face Kizuna Fuhirai.	*Diaries & Daily Life, Gaming*
gr: Μπα σε καλό σου μωρή Ανθουλα μας κοψοχολιασες πάλι ΄σασμός **translation**: Oh my goodness, Anthula, you've cracked us up again #sasmos	Film, TV & Video

Table 3.1: Example of tweets present in each language subset of X-Topic.

Zubiaga et al., 2018; Chua and Banerjee, 2016). Social media platforms like X have become hubs for the exchange of information, opinions, and sentiments, making the development of effective classification methods imperative.

Unsupervised Approaches. Due to the lack of labelled data and the dynamic nature of social media platforms, unsupervised methods have been widely used for topic modelling and classification on the content shared. Several variations of LDA have been introduced that try to address the challenges that arise when working with the often messy and unstructured world of social media. Such solutions, (Zhao et al., 2011; Rosen-Zvi et al., 2004; Steinskog et al., 2017) often try to combine author information with the text shared. Other approaches use unsupervised clustering algorithms, such as k-means or hierarchical clustering, to group similar social media content based on their topic similarity (Wang et al., 2017). These methods are particularly useful when the underlying topics are not predefined and need to be inferred from the data. However, a drawback of these unsupervised approaches is that the derived topics may not always be easily interpretable or comparable across corpora.

Multilingual resources in social media. Supervised methods for topic classification in social media content involve training machine learning models on labelled data. While supervised approaches have demonstrated robust performance on social media tasks (Huang et al., 2013; Camacho-collados et al., 2022), there is a notable scarcity of labelled data for social media content, particularly in languages other than English (Selvaperumal and Suruliandi, 2014); while a lot of the available datasets offer a limited taxonomy of topics (Vadivukarassi et al., 2019). Multilingual and cross-lingual topic classification in social media is therefore a limited explored area. It involves dealing with content in multiple languages, addressing language-specific nuances, and ensuring effective classification. Few resources and models are designed to handle multilingual topic classification. Existing datasets e.g. in Portuguese (Daouadi et al., 2021), Spanish (Imran et al., 2016), Urdu (Kausar et al., 2021) and others (Chowdhury et al., 2020), often suffer from weak labelling or a limited taxonomy of topics, or they are created to solve specific problems e.g. sentiment analysis (Muhammad et al., 2023) and hate speech (Ousidhoum et al., 2019). This presents a gap in the field as many social media platforms have a global user base. Our work addresses this gap by introducing the X-Topic dataset, which includes tweets in four different languages (English, Spanish, Japanese, and Greek), thereby expanding resources for multilingual topic classification in social media.

3.3 X-Topic, a Multilingual Tweet Topic Classification Benchmark

In this section, we describe our methodology to construct, a multilingual tweet topic classification benchmark. First, we describe the original English-based TweetTopicdataset, which we take as inspiration to construct a fully multilingual dataset.

TweetTopic (Antypas et al., 2022) is an English Twitter topic classification dataset consisting of a total of 11,267 English tweets assigned one or more classes from a predefined list of 19 topics such as "News & Social Concern", "Sports", and "Fashion & Style". The taxonomy of topics was decided by a team of social media experts and aims to cover the majority of content being shared in social media platforms. The tweets were distributed over time, from September 2019 to October 2021 and were extracted using keywords of trending topics in each week during the period. Each entry was labelled by five different annotators, and the topic was assigned if there was an agreement of at least two annotators.

In our work, we leverage the taxonomy originally presented in TweetTopic as a foundation for collecting a new set of recent tweets, leading to the introduction of X-Topic. X-Topic is mainly distinguished by its inclusion of entries in four diverse languages: Spanish, Greek, Japanese, and English.

3.3.1 Language Selection and Tweet Collection

The selection of languages was made by taking into account their popularity and practicality. X-Topic is a resource that helps to the analysis of frequently used languages in *X* (English, Spanish, Japanese) as well as a less frequently studied one (Greek). This linguistic diversity also provides a unique opportunity for comparative analysis between linguistically distant groups, such as Japanese and Greek. Moreover, our choice of the September 2021 to August 2022 timeframe continues the timeline of previous work and facilitates engaging in temporal analyses.

For the collection of the dataset, we follow a similar approach to that of the original TweetTopic. Initially, the Twitter API was utilised to collect 50 tweets every two hours for each language. However, in contrast to TweetTopic, we do not use any keyword filtering in our queries. In this way, we acquire a diverse set of tweets, approximately 220,000 tweets for each language, which is closer to the real distribution of content shared in *X*.

3.3.2 Preprocessing

Following the collection of the raw tweets we apply several preprocessing steps. First, we remove potentially remaining tweets in other languages by using a fastText-based language identifier (Bojanowski et al., 2017) on top of the Twitter pre-defined language identifier. Then, we remove tweets that are not in our target period, tweets containing incomplete sentences (too short or end in the middle of the sentence), or abusing words by applying some simple rule-based heuristics. We

also apply a *near-duplication* filter to drop duplicated tweets. This process begins by normalising each tweet (i.e. remove irrelevant substrings and lemmatisation), and then retaining unique tweets only in terms of the normalised form. To ensure the quality of the tweets' content we remove entries that contain URLs, and those where multiple (more than four) emojis or mentions are present.[1] Finally, we sample 1,000 tweets from the remaining set of tweets after preprocessing for each language. The sampling is weighted based on the retweet count of each entry as well as the follower count of the user posting the tweet. This weighting is applied with the assumption that a higher quality content is usually more popular. As a final preprocessing step we mask all mentions of non-verified users with {USER} to ensure the privacy of users.

3.3.3 Annotation

The annotation process closely mirrored the procedure established in TweetTopic. Specifically, each entry of the dataset was annotated by five coders, where each coder had to select one or more labels from a selection of 19 topics in total. A topic was assigned to a tweet only if at least two annotators were in agreement about it. Following previous work on multi-label classification (Mohammad et al., 2018), we refrained from utilising a majority rule in order to create a more realistic and challenging dataset.

The coders who worked on this task were selected and filtered through the *Prolific.co* platform based on their fluency in the corresponding target language.

[1] Detailed number of tweets dropped in each preprocessing step can be found in Table **??**, Appendix **??**.

The actual annotation was performed through an interface created with qualtrics^{XM}.[2] We did not utilise Amazon Mechanical Turk (AMT) due to both the lack of non-English annotators in AMT, as well as, due to the better quality of annotators present in *Prolific.co*. Finally, we ensured the quality of the annotations as our research team includes native speakers in all the non-English languages, who monitored the whole annotation process for each language.

To assess the quality of our annotation process, we report the following three annotation agreement metrics: (1) Krippendorff's Alpha (*Alpha*) (Krippendorff, 2011), (2) Percent Agreement (*PA*), ratio of number of agreements to the total number of annotations, and (3) Agreement between each pair of coders on at least one label (*Overlap*). When comparing our results with those achieved in the TweetTopic annotation, as presented in Table 3.2, we can observe an overall smaller concordance among coders. The highest *Alpha* score observed was 0.26 in the Greek dataset, in contrast to TweetTopic's 0.34. Nevertheless, the agreement metrics remain on par with similar multi-label annotation tasks such as the datasets *Affect in Tweets*, with a Fleiss' Kappa score of 0.26, (Mohammad et al., 2018) and *GoEmotions* (Demszky et al., 2020), with an *Alpha* score of 0.24, noting that a random annotation process would yield an *Alpha* score of 0.

3.3.4 Descriptive Analysis

X-Topic encompasses a total of 361 distinct topic combinations within its 4,000 tweets, showcasing its diversity in themes and coverage. In Table 3.1, we present illustrative entries from our dataset for each language, displaying various topics.

[2]The annotation guidelines for each language can be found in Appendix **??**.

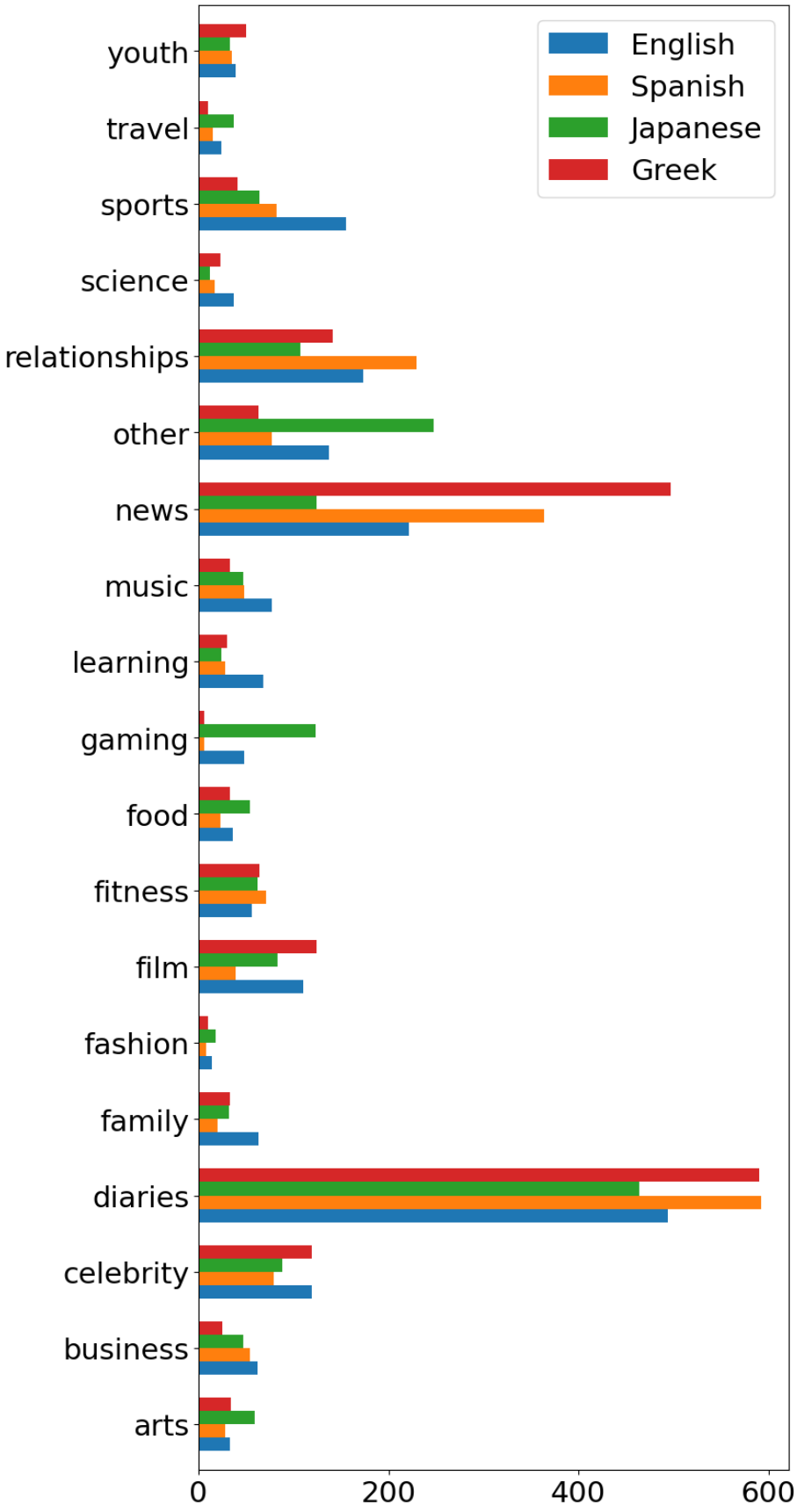

Figure 3.1: Number of tweets per topic and language.

	Alpha	PA	Overlap	AVG Topics
English	0.23	0.87	0.60	2.0
Spanish	0.23	0.89	0.63	1.8
Japanese	0.21	0.87	0.48	1.7
Greek	0.26	0.89	0.74	1.9
TweetTopic	0.34	0.90	0.70	1.6

Table 3.2: Annotator agreement in each language subset of X-Topic and TweetTopic, as well as the average number of topics (*AVG Topics*) assigned to each tweet.

Notably, each tweet, on average, is associated with 1.8 topics, with none of the entries assigned more than 5 topics.

Topic overlap. Upon examining the overlap between topics across all languages, as depicted in Figure **??**, Appendix **??**, we observe interesting patterns. For instance, the *diaries_&_daily_life (diaries)* topic frequently co-occurs with other topics, such as *family* (79%) and *relationships* (76%). Furthermore, there is a substantial overlap between topics that we expected to be closely related in online discussions. For instance, *music* and *celebrities_&_pop_culture* exhibit a 45% overlap, while *youth_&_student_life (youth)* and *learning_&_educational (learning)* demonstrate a 25% overlap.

Topic distribution. As seen in Figure 3.1[3], *diaries_&_daily_life* is the majority class across all four language subsets with 494, 592, 464, and 590 tweets present in English, Spanish, Japanese, and Greek respectively. When looking at less popular topics, differences between languages start becoming apparent with

[3]A map of topic name abbreviations is provided in Appendix **??**.

news_&_social_concern being the second most popular topic for English, Spanish, and Greek (221, 364, and 497 tweets respectively), and *other_hobbies* being the second most popular topic in Japanese (248 tweets). This is in contrast to the TweetTopic dataset which also exhibits an imbalanced distribution but to a lesser degree. This difference can be explained by the fact that in X-Topic we randomly extract tweets from X, aiming to replicate a realistic distribution, rather than utilising trending keywords. These variations in the topic distributions among the four languages, along with differences in the average post length (average number of characters: en: 149.02, es: 128.93, gr: 144.71, ja: 48.58) and the usage of emojis (average number of emojis: en: 0.43, es: 0.42, gr: 0.25, ja: 0.34), provide initial evidence of deeper differences between languages and cultures, present initial evidence into the challenges for developing cross-/multi-lingual models.

3.4 Experimental Setting

In this section, we introduce the models that we evaluate using X-Topic and outline the various settings employed for our analysis.

3.4.1 Data & Settings

To investigate the robustness of our models and the quality of the collected data, we perform a multi-purpose evaluation in a cross-validation setting. For each language subset of X-Topic, we implement a 5-fold cross-validation approach, with each fold encompassing 720/80/200 tweets for the train/validation/test sets. We ensure, whenever possible, that at least one instance of each topic is represented in

each split. Then, we evaluate the following settings in the test splits of X-Topic.

Zero-shot (zero). No training data are provided. This setting aims to investigate the performance of zero-shot and unsupervised systems such as recent instruction tuning (Chung et al., 2022) and generative language models (Bubeck et al., 2023) in low-resource settings.

Few-shot (few). Five entries selected from the validation set of each fold are provided as examples. We aimed to maximise the coverage of topics present when selecting the entries. The goal of this setting is to assess the model's ability to generalise to new tasks or domains with limited training examples. For both the zero and few-shot settings the prompts utilised are similar to the ones used for the training of the BLOOMz and MT0 models in Muennighoff et al. (2022) (see Appendix **??**).

Cross-lingual (TweetTopic)**.** In this setting, we utilise the full English TweetTopic dataset (Antypas et al., 2022) as training set. The goal of the setting is to develop a cross-lingual classifier which will be evaluated on the language-specific test sets of X-Topic. This setting can serve as an indication of the performance in other languages not included in X-Topic for which training data is not available. In addition to the cross-lingual challenge, this setting will have the added temporal challenge, as training and test sets come from different time periods.

Monolingual (target). For each target language, we only make use of its respective training/validation splits in each fold to fine-tune classifiers, which are then evaluated on their respective test sets of the same language. The purpose of this configuration is to assess the capabilities of classifiers across languages as well

as to learn from a limited amount of data.

Multilingual (all languages). In this scenario, we fine-tune a single model utilising all available training data in X-Topic in each fold, aiming to investigate the potential benefits of using a larger amount of training data and the model's capabilities in learning from labelled data in different languages.

For both the monolingual and multilingual settings above, we also explored the setting in which we add the original English TweetTopic as additional training data. The reason for this is to have a setting that includes all training data available, which is a common setting in many NLP tasks in which a larger amount of English data is available.

3.4.2 Comparison Models

We consider two types of models depending on whether they are fine-tuned, or used out of the box in zero- or few-shot settings via prompting.

3.4.2.1 Fine-tuning

We consider five different multilingual models, both general-purpose and specialised on social media and of different sizes, for the fine-tuning setting.

bernice (DeLucia et al., 2022b), a RoBERTa-based model trained on a large corpus of 2.5 billion tweets employing a customised tweet-focused tokenizer. Its training data includes 66 different languages with English, Spanish, and Japanese

being the first, second, and fourth most frequent languages, making it an ideal candidate for the task at hand.

XLM-R (xlmr) (Conneau et al., 2019), a RoBERTa-like model trained on the CommonCrawls corpus (Wenzek et al., 2020) on 100 languages; and **XLM-T (xlmt)** (Barbieri et al., 2022), another XLM-R based model that utilises the last XLM-R checkpoint and further trains on a diverse dataset of over 1 billion tweets spanning over 30 languages.

For models based on XLM-R, we evaluate both the base and large versions. The inclusion of non-social media specific models (*xlmr*) is valuable as it offers insights into their performance in scenarios where the model is not specifically trained on social media content, shedding light on the inherent challenges of such settings. The implementation provided by Hugging Face (Wolf et al., 2020) is used for the fine-tuning of all the models. Hyper-parameter tuning, including batch size, epochs number, learning rate, and weight decay is conducted using Ray Tune (Liaw et al., 2018)[4].

3.4.2.2 Zero and Few-shot

In order to assess the zero/few-shot capabilities of large language models in our task, we compare four models of different sizes and architectures.

BLOOMZ (bloomz) (Muennighoff et al., 2022), a decoder-only model based on the BLOOM models and trained with the xP3 dataset (Scao et al., 2022) with 7 billion parameters.

[4]Details of the models used can be found in Appendix **??**.

mt0 (Muennighoff et al., 2022), a multilingual variant of the multilingual Text-to-Text Transfer Transformer model (Xue et al., 2020). *Mt0*, similarly to *bloomz*, is further trained on the xP3 dataset using multitask prompted finetuning.

chat-gpt-3.5-turbo (chat-gpt) from OpenaAI, [5] an encoder/decoder model with approximately 175 billion parameters (Brown et al., 2020b).

gpt-4o the latest and best performing model from OpenAI which significantly outperforms its predecessors.

3.4.3 Evaluation Metrics

Due to the nature of X-Topic, we use the macro-F1 score, which assigns equal weights to each label, as the evaluation metric. This metric is often used for multi-label classification tasks (Hazaa et al., 2023; Lipton et al., 2014; Mohammad et al., 2018). In order to better understand the performance of the models and due to the imbalanced nature, which can be a challenge for a model's performance evaluation (He and Garcia, 2009), micro-F1 is also reported.

3.5 Analysis of Results

The average macro and micro F1 scores for each model tested across various settings are presented in Table 3.3. Overall, the task presents a challenge for the tested models, with the top-performing classifier, *xlmt-large*, achieving an average

[5] https://openai.com/chatgpt

			English					Spanish					Japanese					Greek				
		TweetTopic	✓			✓	✓	✓			✓	✓	✓			✓	✓	✓			✓	✓
		Target		✓		✓			✓		✓			✓		✓			✓		✓	
		All			✓		✓			✓		✓			✓		✓			✓		✓
macro-F1	finetuned	bernice	55.9	42.7	55.2	58.7	60.3	52.0	26.6	51.5	55.8	55.9	45.8	39.9	55.2	53.3	54.3	41.4	26.4	40.1	43.4	44.0
		xlmr_base	47.0	25.1	45.9	58.0	57.6	42.4	11.6	35.1	48.4	49.1	34.4	2.7	39.9	50.1	52.5	29.5	12.3	34.2	40.0	39.7
		xlmr_large	57.2	51.1	58.7	60.8	**63.3**	51.8	32.6	49.4	53.0	57.2	49.1	38.5	55.9	56.6	56.7	44.0	26.7	45.6	45.5	46.2
		xlmt_base	55.4	42.7	55.1	59.1	60.3	48.5	29.9	49.1	52.8	54.2	47.8	29.5	50.8	53.1	54.4	32.6	21.8	39.6	41.3	45.4
		xlmt_large	60.2	52.0	59.9	62.1	61.7	52.9	45.4	54.4	56.6	**60.0**	50.9	50.9	57.3	57.2	**58.5**	40.6	30.1	49.3	48.6	50.3
	zero	bloomz			23.4					15.5					15.2					1.5		
		mt0			34.7					29.2					37.3					24.7		
		chat-gpt			44.9					37.2					35.6					33.2		
		gpt-4o			59.1					52.4					51.9					49.5		
	few	bloomz			21.0					17.3					14.0					5.2		
		mt0			35.7					29.1					39.0					25.1		
		chat-gpt			54.1					43.6					43.9					39.5		
		gpt-4o			60.0					52.8					53.3					**51.0**		
micro-F1	finetuned	bernice	63.5	63.1	67.6	67.1	66.8	64.9	68.2	71.8	72.5	72.4	52.8	55.3	59.7	59.9	59.3	64.4	68.6	71.1	71.9	70.8
		xlmr_base	57.3	51.9	62.6	65.0	64.0	59.5	57.8	68.5	69.4	70.3	43.8	20.1	52.7	55.8	56.4	53.8	60.3	68.1	69.0	67.8
		xlmr_large	64.4	66.2	67.5	67.2	**68.8**	65.4	69.2	71.6	71.7	72.4	52.3	52.5	59.6	59.2	58.6	64.4	68.7	72.6	72.1	71.2
		xlmt_base	63.5	63.5	66.6	66.9	66.2	63.3	68.7	71.7	72.5	71.5	51.8	49.5	57.8	57.5	58.7	58.5	67.0	70.0	69.8	70.1
		xlmt_large	66.3	66.3	68.7	**68.8**	67.8	67.0	72.5	73.9	73.9	**74.5**	56.0	59.6	**61.4**	60.5	61.3	65.8	70.6	**74.5**	73.0	73.4
	zero	bloomz			24.3					15.2					19.3					0.7		
		mt0			38.7					24.7					42.7					43.2		
		chat-gpt			48.6					49.8					39.2					46.6		
		gpt-4o			63.6					65.6					56.6					65.1		
	few	bloomz			23.5					14.6					17.4					4.4		
		mt0			38.8					25.2					41.8					45.5		
		chat-gpt			57.2					54.9					44.3					53.9		
		gpt-4o			63.2					62.3					57.8					68.6		

Table 3.3: F1 scores (macro & micro average) for each setting tested in 5-fold cross validation. Fine-tuned models are evaluated on different settings depending on the used training data. *TweetTopic:* TweetTopic was used for training; *Target:* the respective language subset of X-Topic was used for training; *All:* all language subsets of X-Topic were used. The best result for each language is bolded, and underlined scores indicate statistically significant difference with respect to the second best score.

performance of 57.6% macro-F1 when trained on all available data (TweetTopic and X-Topic). The majority of models demonstrate better micro-F1 scores, as they are not penalised as heavily for errors in less frequent topics.

3.5.1 Setting Comparison

Cross-lingual capabilities. We analyse the cross-lingual capabilities by comparing the performance of models trained exclusively on TweetTopic with

those trained solely on *Target*, taking only Spanish, Japanese and Greek into consideration. A distinct pattern emerges where cross-lingual models perform competitively (a macro-F1 score of 51.1 for the best model *xlmt_large* on average) consistently outperform their mono-lingual counterparts. For instance, the *xlmr_base* model shows a performance drop of up to 31 points in macro-F1 when tested on Japanese. On average, mono-lingual models display a performance decline of approximately 15 points when compared to their cross-lingual variants. This result is encouraging as it means that cross-lingual models may be used in languages for which training data is currently not available. Even though the models' cross-lingual capabilities are remarkable, it is worth noting that the smaller size of training data available on *Target* (800 instances compared to the 11,267 instances in TweetTopic) has a positive effect on their performance.

Multilingual vs Monolingual. The experiments reveal a consistent increase in performance for multilingual models trained on the entire X-Topic compared to their monolingual counterparts. On average, multilingual models achieve a 17-point improvement in macro-F1. The most significant performance boost is observed in non-English languages, with an average macro-F1 increase of approximately 18 points for Spanish, Japanese, and Greek, compared to only 12 points for the English subset. In general, we observe that cross-lingual models tend to improve as more languages are added. Performance consistently increases with the inclusion of additional target language data or by incorporating more languages. The this trend can bee seen clearly when looking at the overall best-performing model *xlmt_large*, Figure **??**, Appendix **??**.

Zero- and Few-Shot. In both zero- and few-shot settings, when considering macro-F1, *bloomz*, *chat-gpt*, and *gtp-4o* perform better in English and display a

noticeable decline in other languages. In general, *gpt-4o* consistently surprasses the smaller *bloomz7b* and *mt0*, and it's predecessor *chat-gpt*, across all language and metrics. It is interesting to note the differences in performance tha arise in the zero and few-shot benchmarks. The performance of most models, according to macro-F1, increase in the *few-shot* benchmark, *bloomz* being an exception and experiencing a drop of 2.4 points when tested in English. In contrast, *gpt-4o* displays a decrease in micro-F1 scores across all languages indicating a consistent difficulty in maintaining performance when handling imbalanced datasets with more frequent classes.

3.5.2 Model Comparison

Training Corpora. Overall, models trained on X data, consistently outperform the generic *XLMR* models. Notably, both *bernice* and *xlmt_base* demonstrate superior performance compared to *xlmr_base* across all settings and languages, with an average increase in macro-F1 of 11.7 and 8.3 points, respectively. This trend also appears in the larger versions, where *xlmt_large* surpasses *xlmr_large* by an average of 3 macro-F1 points across settings. The performance gap between specific X models and generic *XLMR* models widens in settings with limited training data (trained only on *Target*). Specifically, the X-specific models outperform the generic ones by a significant margin, reaching up to a 37-point increase in macro-F1 (e.g., *bernice* trained on Japanese only) for the base versions and a 12-point increase for the larger versions (e.g., *xlmt_large* trained on Spanish only). These results highlight the benefit of training models on specific domain data.

Fine-tuned models vs few-shot LLM. The experimental results of LLMs reveal that the task is challenging even for larger models. When compared to the

LN	xlmt_large	gpt-4o
en	learning & educational, 78 arts & culture, 76 other hobbies, 74	other hobbies, 85 learning & learning & educational, 73 youth & and student life, 69
ja	news & social concern, 66 business & entrepreneurs, 64 arts & culture, 59	business & entrepreneurs, 84 arts & culture, 76 relationships, 74
es	other hobbies, 83 arts & culture, 68 travel & adventure, 67	other hobbies, 82 youth & and student life, 80 business & entrepreneurs, 75
gr	other hobbies, 89 youth & and student life, 86 arts & culture, 76	other hobbies, 95 youth & and student life, 87 science & technology, 71

Table 3.4: Topics with the highest occurrences of False Negatives errors (topic, error %). The results of *xlmt-large* when trained on TweetTopic and *All*, and of *gpt-4o* in the *few-shot* setting are displayed.

finetuned models, the best performing LLM, *gpt-4o* in the *few-shot* setting, achieves comparable results with *xlmt_base* when fine-tuned on all available datasets, with average macro-F1 of 54.3 and 53.6 for *gpt-4o* and *xlmt_base* respectively, however it achieves the best macro-F1 performance in Greek across all models. In order to better understand the behaviour of each type of model, Table 3.5 displays the average macro Recall and Precision scores achieved by four models of different architectures. Notably, *chat-gpt* seems to struggle more with identifying correctly the assigned labels, as it achieves relatively smaller Precision scores compared to other models. Instead, recall values of *chat-gpt* are similar or higher than other models, particularly for English and Spanish. On average, *chat-gpt* predicts 2, 2.5, 1.5, and 1.4 labels per tweet in English, Spanish, Japanese and Greek, respectively. In contrast, the best performing finetuned model, *xlmt_large*, predicts a more consistent average of 1.7, 1.7, 1.7, and 1.8 labels per tweet on the same languages.

3.5.3 Error Analysis

Using the best overall performing models, *xlmt-large* trained on TweetTopic and *All languages*, and *gpt-4o* in a *few-shot* setting, we attempt to identify patterns in the topics which it struggles the most. Generally, both models attain relatively low recall values (Table 3.5) compared to precision. We analyse this behaviour by examining the topics with the highest occurrences of errors by analysing the False Negative rates (Table 3.4). It is interesting to note the high occurrences of errors noted on the *xlmt_large* results across all languages within the relatively infrequent *Arts & Culture* topic, with error rates of 76%, 59%, 68%, and 76% for English, Japanese, Spanish, and Greek, respectively. In contrast, *gpt-4o* appears to struggle more with the *Youth & Student Life* topic.

Investigating the models' performance in more detail (Tables **??** and **??**, Appendix **??**), reveals a significant weaknesses for both *xlmt_large* and *gpt-4o* in the *Other Hobbies* category. Both models exhibit low performance in all languages with *xlmt_large* and *gpt-4o* achieving 28% and 25% average F1 respectively, highlighting the difficulty in classifying diverse and less defined subjects.

When looking at examples where the models tend to struggle more, there are clear errors like the tweet '*Being on the other side of the casting table today was so much fun. Saying "just have fun with it" and seeing actors literally just have fun with it was amazin*' being classified by *gpt-4o* as "Family" but also there are entries such as *"what are the best web3/crypto newsletters out there not many people know about?"* which is labelled as "News & Social Concern", "Science & Technology" by *xlmt_large* instead of "News & Social Concern", "Business & Entrepreneurs", an arguably valid classification. This behaviour illustrates the difficulty of the task for

	Precision				**Recall**			
	En	Es	Ja	Gr	En	Es	Ja	Gr
chat-gpt	53.0	39.5	46.5	44.0	**63.4**	**63.0**	49.6	43.0
gpt-4o	67.6	61.2	60.8	**63.0**	58.2	53.4	52.6	47.6
bernice	65.9	61.9	57.6	50.0	58.8	56.3	54.5	43.1
xlm_t	**69.2**	**67.7**	**62.1**	61.1	58.1	57.9	**58.4**	**48.2**

Table 3.5: Average macro Precision and Recall scores. Results from the few-shot setting are considered for *chat-gpt* and *gpt-4o*. For the *bernice* and *xlm_t* results we considered models trained on TweetTopic and X-Topic

both human annotators and language models.

3.6 Summary

The aim of this chapter is to expand the resources available for the task of tweet classification, particularly in a multi-label setting and across multiple languages. We introduce the new X-Topic dataset, which includes tweets in English, Spanish, Japanese, and Greek, and is centred around a taxonomy of 19 social media topics. This dataset addresses the lack of labelled multilingual *X* data and encourages the development of new methods for multilingual topic classification.

We explore different model architectures and experimental settings, including zero-shot, monolingual, cross-lingual, and multilingual approaches, to tackle the challenge of multilingual topic classification in social media. Our findings indicate that the task is challenging, especially for less-resourced languages, and that models perform better when trained on a combination of data in various languages. Importantly, our analysis shows how recent LLMs underperform in few-shot

settings in comparison to more efficient but fully-trained multilingual masked language models. Further research should focus on addressing these challenges and enhancing the performance of models in a cross-lingual and multilingual context, for which X-Topic can contribute to as a reliable benchmark.

3.7 Limitations

In this paper, we introduce a valuable new resource that is expected to benefit a wide range of researchers and industry professionals. It is important to acknowledge that there may be differing opinions regarding the methodology used for aggregating the data in X-Topic, specifically the requirement for two annotators' agreement. In any case, we plan to release all the collected annotations, along with the dataset version used in our experiments, to facilitate transparency and further research. The number of languages included in X-Topic selected is relatively small given budget constraints.

Finally, it is important to highlight that while we provide a comprehensive analysis of the cross-/multi-lingual capabilities of five different models, substantial research opportunities remain in exploring the potential of alternative classifiers. This includes investigating the performance and fine-tuning of larger models, considering diverse architectures, and optimising the prompts used for one-shot and few-shot learning.

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